Pre Calc: HW 26 pps 147-148
#1-4 all, 6-11 all, 13-16 all, 43-49 all

Section 1.6 — Transformations

1. $y = x^2 - 3$
Vertical Translation, shift down 3.

2. $y = x^2 + 5.2$
Vertical Translation, shift up 5.2.

3. $y = (x + 4)^2$
Horizontal Translation, shift left 4.

4. $y = (x - 3)^2$
Horizontal Translation, shift right 3.

6. $y = x^2 - 100$
Vertical Translation, shift down 100.

7. $y = (x - 1)^2 + 3$
Horizontal Translation, shift right 1; Vertical Translation, shift up 3.

8. $y = (x + 50)^2 - 279$
Horizontal Translation, shift left 50; Vertical Translation, shift down 279.

9. $y = -\sqrt{x}$
Reflection across the x-axis.

10. $y = \sqrt{x - 5}$
Horizontal Translation, shift right 5.

11. $y = \sqrt{-x}$
Reflection across the y-axis.

13. $y = 2x^3$
Vertical Stretch by a factor of 2.

14. $y = (2x)^3$
Horizontal Shrink by a factor of $\frac{1}{2}$.

15. $y = (0.2x)^3$
Horizontal Stretch by a factor of 5.

16. $y = 0.3x^3$
Vertical Shrink by a factor of 0.3.

Problems 43–46: Function families and transformations

43. $y = 2(x - 3)^2 - 4$

Family: Quadratic Function

Vertical stretch by a factor of 2, shift right 3 units, shift down 4 units.

44. $y = -3\sqrt{x + 1}$

Family: Square Root Function

Shift left 1 unit, reflect across the x-axis, vertical stretch by a factor of 3.

45. $y = (3x)^2 - 4$

Family: Quadratic Function

Horizontal shrink by a factor of 1/3, shift down 4 units.

46. $y = -2|x + 4| + 1$

Family: Absolute Value Function

Shift left 4 units, reflect across the x-axis, vertical stretch by a factor of 2, shift up 1 unit.

Problems 47–49: Write the equation

47. Vertical stretch by 3, then shift right 4 units.

$$\boxed{y = 3(x - 4)^2}$$

48. Shift right 4 units, then vertical stretch by 3.

$$\boxed{y = 3(x - 4)^2}$$

49. Shift left 2 units, vertical stretch by 2, then shift down 4 units.

$$\boxed{y = 2|x + 2| - 4}$$